

VISHWAKARMA AWARDS 2026 • THEME 3

# InspectAI: Low-Cost Defect Inspection for MSMEs

AI-powered surface defect detection on an edge device  
(Raspberry Pi)

**Team Member 1:** Lipi Bagarti

**Team Member 2:** Kartik Ranjan Singh



# Project at a glance

## PROJECT / PRODUCT

InspectAI: Low-Cost Edge-AI Defect Inspection System for MSMEs

## THEME

Theme 3: AI in Hardware & Manufacturing

## SUB-THEME

Computer Vision & Smart Quality Inspection

## SDG ALIGNMENT

SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure)

## ANRF ALIGNMENT

Advanced Manufacturing, AI & Robotics, Industry 4.0/5.0 Technologies



TEAM MEMBER 1

**Lipi Bagarti**



TEAM MEMBER 2

**Kartik Ranjan Singh**

# The problem

## PROBLEM IN ONE STATEMENT

Small factories inspect parts by eye — slow, inconsistent and unrecorded — because commercial machine-vision systems cost lakhs of rupees.



### Slow and tiring

Human eyes tire; defects slip through



### Inconsistent

Calls vary from inspector to inspector



### No data trail

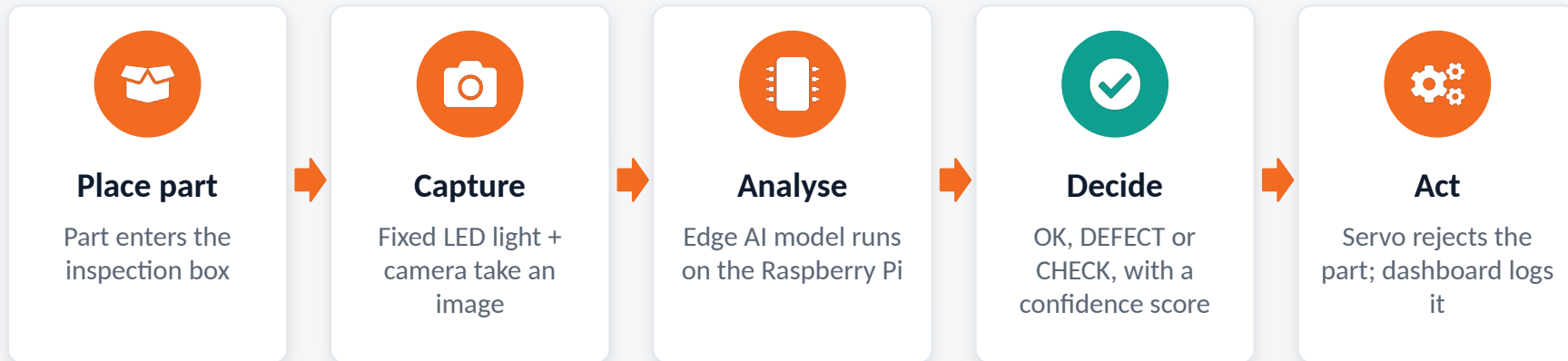
Defects are not logged or analysed



### Too costly to automate

Vision systems are priced for big plants

# An AI inspection box



## Edge AI, works offline

No cloud or internet needed on the shop floor

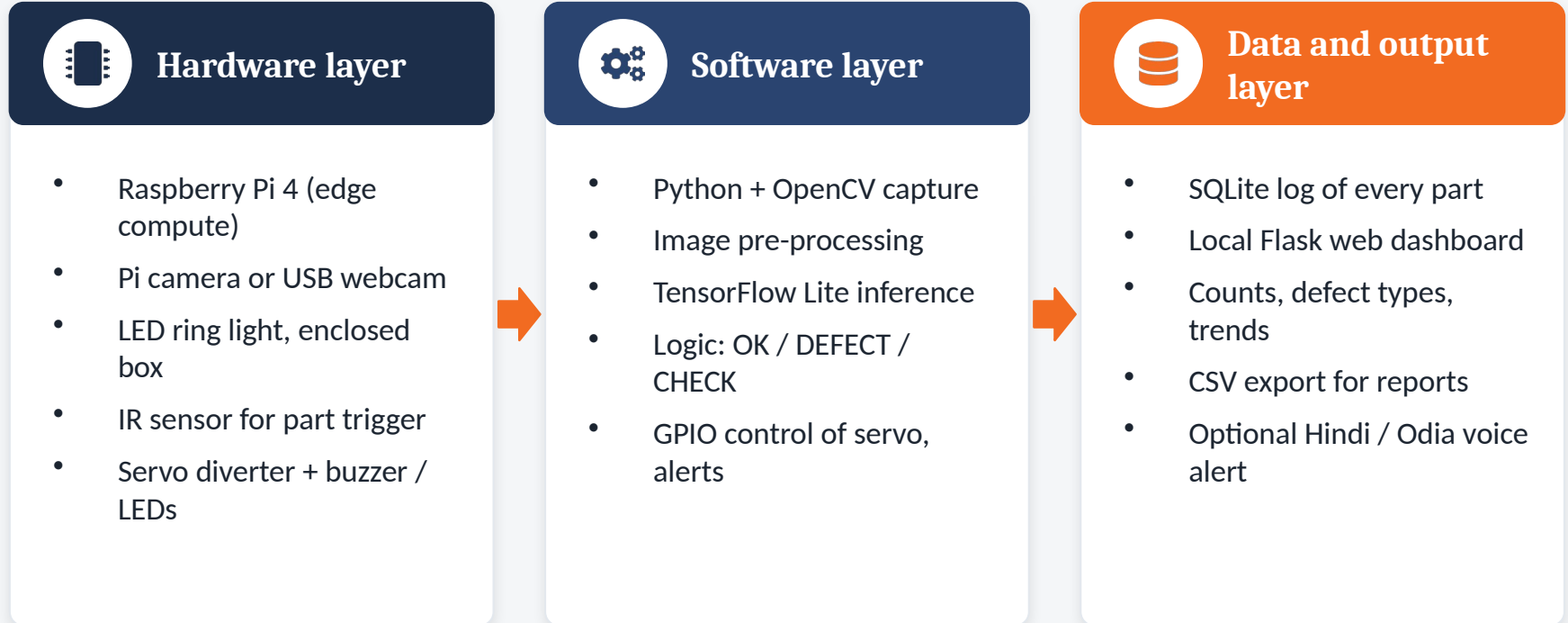
## Real-time decisions

Target: 5+ frames per second on a Raspberry Pi

## Live count and defect log

Simple dashboard: OK / defect count, type and time

# System architecture



# Two models, one decision

## 1. Supervised classifier

*Tells us which defect it is*

- MobileNetV2 or YOLOv8n with transfer learning
- Trained on NEU-DET (6 classes, 1,800 images) plus our own photos
- Classes: crazing, inclusion, patches, pitted surface, rolled-in scale, scratches
- Exported to TensorFlow Lite for the Pi

## 2. Anomaly detection

*Our differentiator*

- Autoencoder or PatchCore trained on good parts only
- Anomaly score = how far a part is from "normal"
- Catches new defect types never seen in training
- Works for factories that have no defect data

# From data to decision



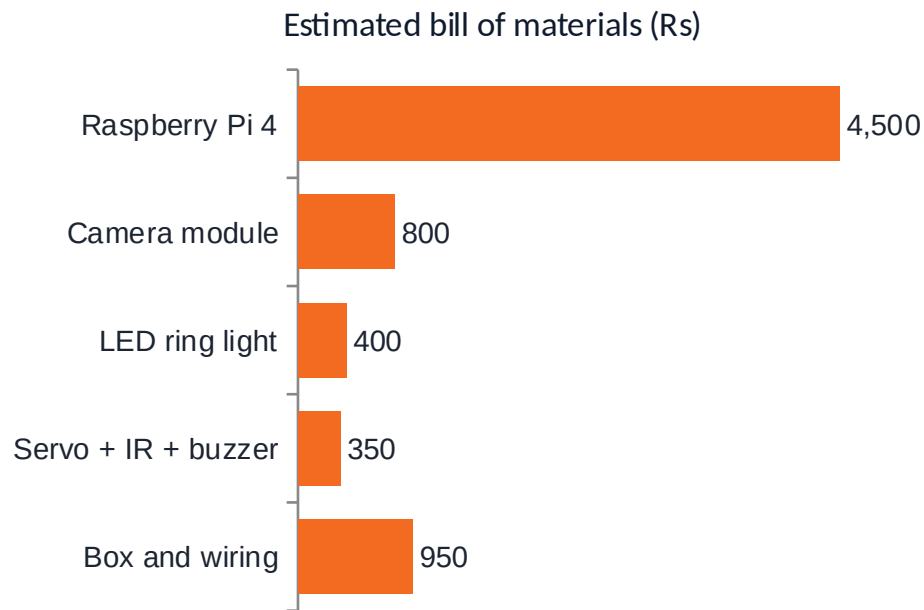
## DECISION LOGIC

| Model result                   | Action                             |
|--------------------------------|------------------------------------|
| Confident DEFECT               | Servo rejects part, event logged   |
| Uncertain (between thresholds) | Routed to a human for manual check |
| Confident OK                   | Part passes, count logged          |

## HOW WE EVALUATE

- Train / validation / test split of 70 / 15 / 15
- Accuracy, precision, recall, F1
- False-reject rate and inference time
- Re-test on our own photos under changed lighting

## Low-cost by design



**~ Rs 7,000**

estimated prototype cost

Commercial machine-vision systems typically run into lakhs of rupees.

*Costs are approximate and will be confirmed at purchase.*

# Who benefits, what we deliver



## Target beneficiaries

- MSME workshops, small foundries and fabrication units
- Quality-control inspectors and supervisors
- Small manufacturers who cannot afford lakh-range vision systems



## Deliverables

- Working InspectAI prototype (box, camera, Pi, servo)
- Trained model files and dataset notes
- Local dashboard and CSV reports
- Test report and setup guide



## Expected outcomes

- Faster, more consistent inspection
- Fewer defective parts reaching customers
- A defect data trail for process improvement
- Lower inspection cost for small units

# Build plan and real-world pilot

W1

## Data + baseline model

Get NEU-DET, train the classifier, set up the anomaly model

W2

## Box + our own photos

Build the light box, mount the camera, collect 100-200 real images

W3

## Deploy on the Pi

Convert to TFLite, wire IR trigger, servo diverter and dashboard

W4

## Test + report

Measure accuracy, false-reject rate and speed; prepare demo

## REAL-WORLD PILOT

1

### Find a partner

A local foundry or fabrication unit for a trial

2

### Run side by side

Compare InspectAI with manual inspection

3

### Refine and hand over

Tune thresholds, train staff, collect feedback

# Team and responsibilities



TEAM MEMBER 1

**Lipi Bagarti**

- Dataset preparation and augmentation
- Model training (classifier + anomaly)
- Evaluation and threshold tuning



TEAM MEMBER 2

**Kartik Ranjan Singh**

- Hardware assembly: box, camera, servo
- Raspberry Pi integration and deployment
- Dashboard and data logging

**Shared by both: proposal, testing, documentation and the final demo.**

# How the solution stays safe



## Fail-safe by design

Uncertain parts go to a human; they are never silently passed



## Less manual handling

An automatic diverter sorts rejects, so inspectors touch fewer parts



## Less eye strain

A clear verdict replaces hours of staring at surfaces



## Electrically safe

Low-voltage components inside an enclosed box



## Guarded moving part

Only the small reject servo moves, kept behind a guard



## Data stays local

Images are processed and stored on the device, with no cloud upload

# How it improves efficiency

**< 1 s**

decision per part

**5+ FPS**

on a Raspberry Pi

**90%+**

target accuracy on test set

**~Rs 7k**

prototype cost

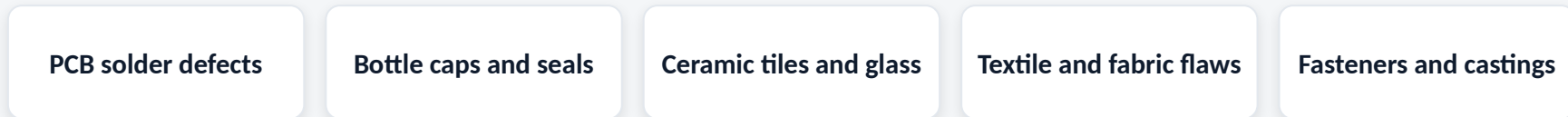
## DESIGN TARGETS, TO BE VALIDATED IN TESTING

- Automatic reject diversion replaces manual sorting of good and bad parts (material handling)
- The same decision quality on every part, all shift, with no fatigue drop
- A live defect log lets supervisors fix the process sooner, so fewer bad batches
- Inspectors are freed for higher-value tasks such as root-cause checks

# Grow in three phases



ADAPTS TO NEW USE CASES BY RETRAINING THE MODEL



# Why this is different

|                       | Manual inspection | Commercial vision | InspectAI          |
|-----------------------|-------------------|-------------------|--------------------|
| Upfront cost          | Low               | High (lakhs)      | Low (~Rs 7k)       |
| Consistent, all shift | No                | Yes               | Yes                |
| Works offline         | n/a               | Vendor-dependent  | Yes (edge AI)      |
| Needs defect samples  | n/a               | Usually yes       | No (anomaly model) |
| Logs defect data      | No                | Yes               | Yes                |

## Learns "good" only

Catches unseen defects without defect samples

## Built for MSMEs

Under ~Rs 10k, offline, simple dashboard

## Swap-a-model design

Same box adapts to new parts by retraining

# Before and after: the inspector's journey



Part arrives



Inspection



Decision



Record

## BEFORE

Parts pile up at the inspection table

Inspector checks each part by eye, hour after hour

Judgement varies; small defects get missed

Rarely written down; problems found late

## AFTER

Part is placed under the camera in the light box

Camera + AI check in under a second (target)

Clear OK / DEFECT / CHECK verdict; rejects are diverted

Every part logged; dashboard shows defect trends

# AI quality inspection, within reach of every small factory.

Low cost: ~Rs 7k

Edge AI: works  
offline

Anomaly detection:  
no defect samples



Thank you

Lipi Bagarti · Kartik Ranjan Singh